

InStore Optima — Development Changes

A summary of all features, fixes, and improvements made in this sprint.

1. Structured Logging (Serilog)

Problem: Default [ASP.NET](#) logs were noisy and hard to read — 5–10 lines per request with full namespace noise.

Changes:

- Replaced the default console output with a clean custom template: `[HH:mm:ss LVL] Message`
- Used `AnsiConsoleTheme.Code` for color-coded log levels (blue INFO, yellow WARN, red ERR) in the terminal
- Added `UseSerilogRequestLogging()` — collapses all [ASP.NET](#) routing/middleware logs into **one timed line per request**: `HTTP GET /api/products responded 200 in 14ms`
- Silenced polling noise (`/api/auth/ping` , `/api/messages/unread-count` , `/api/badges`) — logged at Verbose so they never appear
- Minimum level overrides suppress `Microsoft.*` and `System.*` framework logs at Warning+; only app-level logs show at Information+
- Added a **global** `CrudLoggingFilter` (action filter) that automatically logs every POST/PUT/PATCH/DELETE across all controllers with the acting user's name:
`Alice Created Products via CreateProduct`
- Added explicit login/register logging in `AuthController` with user name, email, and role
- `AuthController` excluded from the CRUD filter (it logs its own events)

Before:

```
[23:58:29 INF] Executing endpoint 'AuthController.Ping'
[23:58:29 INF] Route matched with {action = "Ping"}
[23:58:29 INF] Executing action method...
[23:58:29 INF] Executed action...
[23:58:29 INF] Executed endpoint...
```

After:

```
[23:58:29 INF] HTTP GET /api/auth/ping responded 200 in 3ms
[00:34:11 INF] User ADMIN (Admin@instore.com) logged in with role Admin
[00:34:36 INF] ADMIN Created Orders via CreateOrder
```

2. Automated PO Delivery Notifications (Internal Messaging)

Feature: When a Purchase Order is created, all Managers and Staff automatically receive an internal message from the supplier's dummy email address with a one-click "Mark as Delivered" button.

How it works:

1. Admin/Manager creates a PO with an expected delivery date
2. `PONotificationService` sends one internal message per Manager and Staff user (excluding the creator)
3. The message appears to come from the supplier (e.g., `acmecorp@instore.com`) — stored via `SenderDisplayName` and `SenderDisplayEmail` override fields
4. The message body contains full PO details: PO#, supplier name, expected delivery date, replenishment order#
5. A yellow action bar appears in the message detail with a **"Mark PO #X as Delivered"** button
6. Clicking the button calls `PUT /api/purchaseorder/{id}` → status set to "Delivered" → stock automatically updated, IN movement recorded, GRN generated, replenishment marked Fulfilled
7. Button changes to  **Marked as Delivered** immediately — no page reload

DB changes (migrations added):

- `InternalMessage.ActionType` (string?) — e.g., `"MARK_PO_DELIVERED"`
- `InternalMessage.ActionPayload` (string?) — e.g., `"5"` (the PO ID)
- `InternalMessage.SenderDisplayName` (string?) — overrides sender name display
- `InternalMessage.SenderDisplayEmail` (string?) — overrides sender role/email display

Guards added:

- A PO already marked "Delivered" throws `409 Conflict` — prevents double-delivery and double stock updates
- Frontend shows "Already Delivered" state if another user already acted on the same message

3. Stock & PO Delivery Bug Fixes

Bug 1 — Double delivery: Marking a PO as delivered from multiple messages (different users) would add stock multiple times.

- **Fix:** `UpdatePOStatusAsync` now checks `if (po.Status == "Delivered")` at the start and throws `InvalidOperationException` before touching stock

Bug 2 — Missing stock row: Products created before the auto-create-stock logic was added had no stock row. PO delivery would record the IN movement but silently skip the stock update.

- **Fix:** `UpdatePOStatusAsync` now creates the `Stock` row if it doesn't exist (instead of skipping)

4. Date Picker Timezone Fix

Problem: `new Date().toISOString()` returns UTC time. Users in IST (UTC+5:30) could select yesterday's date because the UTC date was behind the local date.

Fix: Replaced `toISOString().split('T')[0]` with `toLocaleDateString('en-CA')` everywhere — `en-CA` locale always produces `YYYY-MM-DD` in the user's **local** timezone.

Files fixed: `PurchaseOrders.jsx` , `Replenishment.jsx` (both the `min` attribute and the validation logic)

5. Action-Required Badge System

Feature: Live badge counts on the sidebar and top navbar that show staff and managers exactly what needs attention when they log in — similar to unread message badges.

What each badge counts

Location	Badge	Counts
Inventory → Stock		Products where <code>currentStock < minStock</code>
Procurement → Replenishment		Replenishment orders with status <code>Pending</code> (need approval)

Location	Badge	Counts
Procurement → Purchase Orders		POs with status <code>Pending</code> (need delivery confirmation)
Finance → Orders		Orders with status <code>Pending</code> or <code>Processing</code>
Finance → Payments		Payments with status <code>Pending</code> (need to mark completed)
Section headers (Inventory / Procurement / Finance)		Sum of their child link counts

Architecture

- **Backend:** `GET /api/badges` — single endpoint returning all counts via `BadgeCountsDto`
- **Frontend:** `AlertBadgesContext` — React context that polls `/api/badges` every 60 seconds and exposes counts + section totals to the entire app
- **Sidebar (`InnerLayout`):** Badges on individual links (red) and section headers (amber total)
- **Navbar (`TopNavbar`):** Badges on each section pill

Live updates (no page refresh needed)

`fetchBadges()` is called immediately after any action that changes a count:

Page	Trigger
Stock	Update stock quantity, Create stock record
Replenishment	Approve or Reject an order
Purchase Orders	Mark as Delivered or Cancelled
Messages	Mark PO as Delivered via action button
Orders	Update order status
Payments	Mark payment as Completed

Glow + Sound reminder

If a badge stays non-zero for **3 minutes** without action:

- The navbar pill starts a **red pulse glow** animation
- A gentle **two-tone chime** (A5 → E5, generated via Web Audio API — no audio file required) plays every 8 seconds
- **Clicking the glowing navbar pill** silences the sound immediately (glow remains until count reaches zero)
- When the count finally drops to zero (issue resolved), both the glow and sound stop and the timer resets
- The AudioContext is initialised on first user click to comply with browser autoplay policies

Files Changed Summary

Backend (C#)

File	Change
Program.cs	Serilog config, CRUD filter registration, PNotificationService registration, badge polling silenced
InternalMessage.cs	Added ActionType , ActionPayload , SenderDisplayName , SenderDisplayEmail
InternalMessageDTO.cs	Added new fields to MessageResponseDto
InternalMessageController.cs	Maps new fields in ToDto ; sender display overrides sender name/role
PurchaseOrderController.cs	Calls PNotificationService after PO creation; 409 guard on double-delivery
PurchaseOrderRepository.cs	Guards double-delivery; creates missing stock row on delivery
AuthController.cs	Added ILogger , explicit login/register/failed-login log lines
BadgeController.cs	(new) GET /api/badges endpoint
BadgeCountsDto.cs	(new) DTO for badge counts
IPNotificationService.cs	(new) Interface
PNotificationService.cs	(new) Sends internal messages to Managers & Staff on PO creation

File	Change
CrudLoggingFilter.cs	(new) Global action filter — logs all mutations with acting user
appsettings.json	Serilog MinimumLevel overrides
Migrations	AddActionFieldsToInternalMessage , AddSenderDisplayToInternalMessage

Frontend (React)

File	Change
AlertBadgesContext.jsx	(new) Polling context, glow timers, Web Audio chime, muteSection
App.jsx	Registered AlertBadgesProvider
InnerLayout.jsx	Sidebar badges with glow class
TopNavbar.jsx	Navbar pill badges, glow animation, muteSection on click
Messages.jsx	PO action bar with Mark as Delivered button; live badge refresh
Stock.jsx	fetchBadges() after update/create
Replenishment.jsx	fetchBadges() after approve/reject; timezone fix on date picker
PurchaseOrders.jsx	fetchBadges() after status update; timezone fix on date picker
Orders.jsx	fetchBadges() after status update
Payments.jsx	fetchBadges() after mark completed
index.css	.isb-badge , .tnav-alert-badge , .tnav-pill-glow , pulse animation keyframes